

Instructor: Dr. Ching-yu (Austin) Huang

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Mon, Tue, Wed, Thur: 10:00–11:15 am

Class: **Mon, Wed (14:15 - 15:30 pm)**, location: **GLAB 207**Prerequisites: **Admission to CIS Graduate Program** Students without prerequisite(s) must withdraw from the class.

WK	Date		Topic	Reading	Notes	Set
1	9/2/26	Wed	Intro of DB, XAMPP, MySQL, import data	E&N: Ch 1	Install XAMPP	1
2	9/7/26	Mon	Labor Day		No class	
	9/9/26	Wed	SQL DML select, create view, table , eval. test	E&N: Ch 2, 6	Start HW	2
3	9/14/26	Mon	Client-server, web DB app, basic SQL, PHP	E&N: 11, PHP: 9,11	Start Team	3
	9/16/26	Wed	SQL, authentication, PHP MySQL, SQL injection	E&N: 6,30, PHP MySQL		4
4	9/21/26	Mon	DB lifecycle, design, ER model	E&N: Ch 3		5
	9/23/26	Wed	Relational model, constraints, ERD to tables	E&N: Ch 5		6
5	9/28/26	Mon	DDL - create, DML- insert, update, delete & PHP	E&N: Ch 6		7
	9/30/26	Wed	FK, change, cascade, alter, workbench	E&N: Ch 7		8
6	10/5/26	Mon	Complex SQL Queries, set, subquery	E&N: Ch 7		9
	10/7/26	Wed	View, updatable & materialized views	E&N: Ch 7		10
7	10/12/26	Mon	Midterm 1 (slide sets # 1 – 9)	Midterm 1		11
	10/14/26	Wed	Team project presentation – design and tables		Design & features due	12
8	10/19/26	Mon	SQL variables, SQL programming (PL/SQL)	E&N: Ch 10		13
	10/21/26	Wed	SQL stored routines (PL/SQL)	E&N: Ch 10		14
9	10/26/26	Mon	SQL error handling, SQL Triggers	E&N: Ch 10		15
	10/28/26	Wed	Multiple web items, image, GIS	PHP: 16, 17, 23		16
10	11/2/26	Mon	Transactions	E&N: Ch 21		17
	11/4/26	Wed	Concurrency control	E&N: Ch 24		18
11	11/9/26	Mon	Relational algebra, relational calculus & SQL	E&N: Ch 8		19
	11/11/26	Wed	Midterm 2 (slide sets # 1 – 19)	Midterm 2		20
12	11/16/26	Mon	Last day to withdraw with 'W' grade		0% refund	
12	11/16/26	Mon	Functional dependencies & Normalization	E&N: Ch 9, 14	HW due (11/14, Sat)	
	11/18/26	Wed	Emerging DB app – JSON & NoSQL	E&N: Ch 24		21
13	11/23/26	Mon	Data mining, warehouse, OLAP, visualization	E&N: Ch 28, 29		22
	11/25/26	Wed	Filesystem, database storage, indexing	E&N: Ch 16		23
14	11/30/26	Mon	RAID, Indexing, query processing	E&N: Ch 17,18		24
	12/2/26	Wed	Query optimization, SQL performance tuning	E&N: Ch 19		25
15	12/7/26	Mon	DB admin, DCL, Integrity & security	E&N: Ch 30		26
	12/9/26	Wed	School follow Tuesday schedule		No class	27
16	12/14/26	Mon	Final exam (slide sets # 1 – 27)	Final exam	Wed Follow Tuesday	28
	12/16/26	Wed	Review final, Team project presentation		Implementation due	29
17	12/21/26	Mon	Web hosting, backup			30

Last revision: 8/31/2026. For important dates, please consult the Academic Calendar via the following link:

<https://www.kean.edu/offices/registrars-office/academic-calendar>

Content: Enterprise database management systems; database design; database implementation; storage and indexing; transaction processing concepts; data warehousing, data mining and online analytical processing (OLAP); emerging database applications.

Course Student Learning Outcomes: Upon completion of this course, the student will be able to:

- A. Measure the role of databases in enterprise information systems. Compare and summarize diverse database models.
- B. Design and build web-based applications using database management systems.
- C. Explain the role of a database in data mining and OLAP.
- D. Write and execute SQL statements and implement PL/SQL programs. Illustrate transaction processing concepts.
- E. Summarize the emerging database applications.

Instructional Method

- Slides, assignments, and tool info will be posted on the class google drive that will be sent by the instructor.
- For more information about this course and for course materials, go to <http://blackboard.kean.edu>

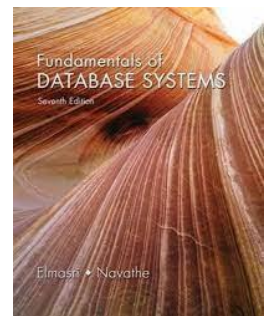
Textbook & Materials

Fundamentals of Database Systems (7th Edition),

Author: Elmasri, R. and Navathe, S.,

Publisher: *Pearson*, 2015.

ISBN-13: 9780133970777



Require hardware and Software

- **This is a hands-on course. Students are required to bring a personal laptop to every class.**
- Database related free software: MySQL Workbench, XAMPP, MySQL
- Other free software: Putty.exe (Win), terminal (Mac), FileZilla, Sublime
- An account on a Linux server will be provided by the instructor.

Open Educational Resources

For more information, please see the Nancy Thompson Library Learning Commons:

<http://libguides.kean.edu/OER>

Assessment and course grading

The course includes Exams, Labs, Homework and Project, a **total of 1000 points**.

- Exams: Midterm1 (200 pts) Midterm2 (200 pts), final (200 pts),
- Team project and presentations (200 pts)
 - 1st presentation: team design
 - 2nd presentation: individual implementation (create tables + PHP programs)
- 1 Homework (200 pts)

The Digital Gradebook (in Canvas) will be used to record student grades.

Grading Policy

A: >= 940				A-: 939 -890											
B+: 889 - 840				B: 839 - 800				B-: 799 – 760							
C+: 759 – 720				C: 719 - 680											
D: 679 – 640															
E: below 640															

B- or better is needed for CIS Graduate program.

Technical Requirements:

1. In order for your Blackboard course to function correctly, you will need to disable pop-ups on your Internet browser.
2. Make sure you have Microsoft Office installed on your computer. You may be eligible for a free MS Office Software Student Edition. You are required to create an account and provide a valid Kean University ID to obtain access to the software applications. To start the application process, go to the [Office 365 Education website](#).

Academic Early Alert information will be provided to students by the instructor via KeanWise. Please see the 2020-2021 Undergraduate or Graduate Catalog for more information.

Class Policies:

- **Posting of course materials:** **Public posting of course materials from this class is prohibited.** Materials presented in the class are for the personal use of registered students during the semester only. Homework, programs, and other classwork should not be shared, distributed or publicized in any way, for the protection of other students and in respect of the intellectual property rights of the faculty and copyright owners. Kean University's Academic Integrity Policy can be found here: <https://www.kean.edu/media/academic-integrity-policy>.
- It is students' responsibilities to verify all the scores posted on **the class website** that is announced on the first day of the class.
- **No late work** will be accepted without permission in advance from the instructor. Late work is penalized.
- **No make-up exams** are provided without documentation of medical or family emergency. Please arrange your schedule for the next 16 weeks to arrive in class on time and prepared.
- **Attendance** and participation are required.
- **All homework assignments, projects, and exams are expected to be completed by the students themselves.** **No credit will be given for an assignment or homework that is copied – in part or in total – from another person. The use of AI tools such as ChatGPT, GitHub Copilot, and similar resources is not permitted for these tasks.**
- **Exam policy:** During class examinations, if a student leaves the classroom for any reason during the exam time, **the exam must be turned in before the student departs, concluding the exam for that student.** The student cannot resume the exam on return later. You may not make or take a phone call, text or access the internet during class or a test.
- **Classes may not be audio or digitally captured**, unless permission has been granted by the instructor for that specific class.

Privacy Statement

Assuring privacy among faculty and students engaged in online instructional activities helps promote open and robust conversations and mitigates concerns that comments made within the context of the class will be shared beyond the classroom. As such, recordings of instructional activities occurring in online classes may be used solely for internal class purposes by the faculty member and students registered for the course, and only during the period in which the course is offered.

Important University Policies and Information

- All students and faculty should familiarize themselves with the **Cougar Restart Guide** which includes important virtual resources for students by Kean University via the following link:
 - <https://www.kean.edu/coronavirus-information/resources-current-prospective-students>
- Students are responsible to review and understand the *University Academic Integrity Policy* available via the following link: <https://www.kean.edu/academic-integrity>
- *Instructors must indicate on the syllabus and/or individual assignment if the use of Kean University academic support services ([Tutoring and/or Writing Center](#)) is permitted for take home assignments/exams.*
- Students should review the *Student Code of Conduct*, as it discusses expectations of appropriate conduct in the classroom:
 - <https://www.kean.edu/offices/community-standards-and-student-conduct/student-code-conduct>
- Students are strongly encouraged to register for the University's emergency notification system (<http://www.kean.edu/campusalert>) in order to be informed of campus emergencies, weather notices, and other announcements. Kean Ocean students would sign up via the following link:
 - <https://ocean.sendwordnow.com/LicensePage.aspx>
- All students must have a valid Kean email account. For those who do not already have one, forms are available on-line at <https://www.kean.edu/offices/computer-and-information-services/ocis-forms>
 - click on E-mail Account Request Form.

Diversity & Non-Discrimination Statement

Kean University is committed to establishing and maintaining a diverse campus community through inclusive excellence and equal opportunity. Kean's commitment to access and equity is designed to prepare each graduate to not only thrive, but climb higher in a diverse world. As an affirmative action, equal opportunity institution we work to support a campus-wide agenda to foster a community that both values and promotes the diversity and equity of all students, faculty, staff, administrators, and beyond. Kean adheres to the [New Jersey State Policy Prohibiting Discrimination in the Workplace](#); for more information on Kean's non-discrimination policy and procedures, and protected categories under the policy, please visit: <https://www.kean.edu/offices/affirmative-action>.

Americans with Disabilities Statement

Students with documented disabilities who may need instructional accommodations or special arrangements in the event of an evacuation should notify the instructor as soon as possible and no later than the second week of the term. Students may also contact The Office of Accessibility Services by phone at (908) 291-3137 or email accessibilityservices@kean.edu to discuss special needs, accommodations and arrangements.

Title IX

Title IX of the Education Amendments of 1972 (Title IX) prohibits discrimination on the basis of sex in education programs or activities. Sexual harassment in any form will not be tolerated at Kean University. Sexual harassment by students should be reported to the Office of Affirmative Action Programs, Office of the Vice President for Student Affairs or the Office of Community Standards and Student Conduct immediately. Information about the University's Sexual Misconduct Policy may be found at the following: <https://www.kean.edu/offices/policies/sexual-misconduct-policy>.

Syllabus Updates:

Instructor reserve the right to change the syllabus at any time throughout the semester. All changes will be announced in class and a new syllabus will be electronically available to all students. If you are enrolled in this class past the add/drop date you are subjected to all rules in this syllabus.